David Pedroza Segoviano

Lead Software Engineer

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PROFESSIONAL EXPERIENCE

PeopleOPTI , *Lead Software Engineer* 11/2023 – Present | Remote, México

Joined as a data algorithm specialist while completing my Physics degree and rapidly advanced to leading complex LLM multi-agent system development for HR analytics clients. Transformed data processing workflows, implemented cloud infrastructure best practices, and pioneered AI-driven insight solutions for enterprise clients including José Cuervo, FEMSA and HDI, establishing new technical standards across the organization.

- Optimized Organizational Network Analysis algorithm, reducing processing time from 4 hours to 10 minutes for 20 million data points (95% improvement)
- Redesigned ETL pipelines, cutting processing time by 98% (from 30 minutes per 200K records to 2 minutes per 1M records)
- Developed cloud-based interactive dashboard with NextJS/FastAPI that decreased execution time by 80% and development time by 70%
- Created agent-based LLM systems using Langchain/Langgraph that reduced insight generation time by 70% through RAG and CAG approaches, using a variety of techniques and technologies such as MCP Server Development, A2A connections, Data Analytics and Data Engineering AI Agents and Memory consistent agents using profiling and DB indexation
- Established technical assessment frameworks resulting in the successful hiring of 2 specialized data/ML team members
- Built ML correlation models that helped a key client (FEMSA) reduce employee turnover by 20%
- Applied Azure cloud best practices within our infrastructure utilizing Web Apps, Container Registry, Azure Functions, Queue Storage, and AML Workspaces with specialized VMs for enterprise-grade data processing, reducing cloud costs up to 50%

Código Facilito , *Instructor and Course Developer* 06/2024 – 03/2025 | Remote, México

Delivered specialized technical instruction for Latin America's leading Spanish-language technology academy, creating and teaching 10+ courses across Data Science, Machine Learning, Data Engineering, and Python Backend bootcamp programs. Developed comprehensive curriculum materials and provided personalized mentorship to 500+ students.

- Created complete course materials including lesson plans, presentations, coding exercises, and assessments for technical topics such as Data Visualization, ETL/ELT, ML Model Deployment, and FastAPI
- Maintained 90% average student completion rate across all programs
- Designed practical projects that improved knowledge retention by 40% over theoretical approaches
- Delivered interactive 2.5-hour live sessions with real-time Q&A and provided continuous support
- Evaluated student projects with detailed feedback to enhance their technical capabilities

Deloitte , *Analyst* 01/2022 – 01/2023 | Remote, México

Designed and applied efficient cloud infrastructure solutions for enterprise clients, focusing on cost optimization, data processing improvements, and workflow automation to drive measurable business outcomes.

- Reduced cloud infrastructure costs by 30% through strategic AWS resource optimization for large-scale organizations
- Enhanced data processing efficiency by 15% by developing streamlined data engineering pipelines
- Automated manual reporting workflows using AWS CLI and bash scripting, saving teams 15+ hours weekly

Codeando México , *Full Stack Developer* 10/2021 – 03/2022 | Remote, México

Led the development of an impactful open-source humanitarian platform that helps users locate missing people, combining technical expertise with social impact to create accessible visualization tools.

- Spearheaded development of the Socorro Bot Web Page , serving 100+ daily users searching for missing persons
- Developed a user-friendly data visualization interface by linking Airtable API, Jekyll, JavaScript, and Leaflet;

EDUCATION

B. Sc. in Physics, *University of Guanajuato* 08/2018 – 12/2023

León Guanajuato, México

- Completed advanced coursework in Machine Learning, Deep Learning, OOP, and Data Analysis with an overall score 91/100 (~3.7/4.0 GPA)
- Completed a capstone project in computational astrophysics, developing a simulation that enhanced understanding of galaxy formations
- Active member of the university's Tech Club, participating in monthly hackathons and coding challenges, honing practical coding and teamwork skills such as effective communication and task delegation

SKILLS

Programming Languages: Python, JavaScript, C, lua

LLM Agentic System Development: Langchain, Langgraph, Langsmith, Model Ccontext Protocol, ChromaDB, advanced pre-retrieval, retrieval and post-retrieval techniques for RAG systems and tool creation for Agents

Frontend: HTML, CSS, ReactJS, Jekyll, Tailwind, Shadcn

Backend: NextJS (Full Stack), FastAPI, Flask, Streamlit, OAuth2

DevOps: Docker, Git, GitHub, GitHub Actions

Database Knowledge: SQL, MongoDB, SQLAlchemy ORM for Python

Microsoft Azure: Azure Functions, Web Apps, Container Apps, Aplication Insights, Billing Monitoring, VMs, Bot Framework for MS Teams, Databases services like SQL Servers and Azure Cosmos DB, Machine Learning and AI Services such as Azure Machine Learning, Azure AI Search, and Azure OpenAI

Other Technical Skills: Neovim btw, Linux Environment

LANGUAGES

Spanish — Native/Bilingual

English — Proficient

ORGANIZATIONS

Clubes de Ciencia Mexico, *Instructor* 08/2019 – Present | Mexico

Clubes de Ciencia México is a non-profit organization that brings graduate students and professionals to teach intensive science workshops in underserved communities across Mexico. I began my involvement as a student participant in 2019, then progressed to volunteer the following year, and subsequently served as a Teaching Assistant. Since 2023, I have been an instructor developing and leading programming-oriented workshops that integrate computational methods with physics, chemistry, and biology concepts. Through this role, I design interdisciplinary curricula that demonstrate how programming and computational thinking can be applied to solve scientific problems, helping to inspire and educate students from diverse backgrounds about the intersection of technology and natural sciences.

GitHub Campus Expert, *Campus Expert* 08/2022 – 08/2024 | Remote

GitHub Campus Expert is a selective program that trains student leaders to build technical communities at their universities and beyond. As a Campus Expert, I organized numerous events and workshops focused on software development, open source technologies, and technical skills development. Through this role, I facilitated knowledge transfer and promoted science and technology accessibility across various universities and communities throughout Mexico, helping to bridge the gap between academic learning and practical industry applications while fostering collaborative learning environments for students and professionals.

Future Lab, *Coordinator* 08/2019 – Present | Remote

Future Lab is an innovation and technology development group focused on fostering digital skills and technological entrepreneurship. As a coordinator in Future Lab, I have conducted workshops on emerging technologies and have collaborated on various technical projects aimed at solving real-world problems.

AWARDS

1st Place, *UST D3CODE Hackathon - Mexico* 09/2024

1st Place, *Nasa Space Apps Challenge - Guanajuato* 11/2023

Gold Medal, *Regional Programming Olympiad of Guanajuato* 11/2017

Gold Medal, 05/2017
Regional Hight School Mathematical Contest - Fedematicas

1st Place, 05/2017
State Physics Olympiad of Universidad de La Salle Bajío

2nd Place, *World Educational Robot Contest - Mexico* 10/2016

2nd Place in Brick Robot Contest, 11/2016
World Educational Robot Contest - China

2nd Place, *Lincehack - State Hackathon* 09/2019

3rd Place, *HackLeon - State Hackathon* 04/2019

Bronze Medal, *Regional Programming Olympiad of Guanajuato* 11/2016

Honorable mention, *Regional Physics Olympiad of Guanajuato* 06/2017

Honorable mention, 04/2017
Regional Mathematics Olympiad of Guanajuato

Honorable mention, 10/2016
Regional Chemistry Olympiad of Guanajuato

Honorable mention, 05/2017
State Robotics Contest of Universidad de La SAlle Bajío